Hints: Polynomials – 5. Integer coefficients, divisibility and irreducibility

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Gauss: if $g, h \in \mathbb{Z}[x]$ are primitive and a prime p divided all coefficients of gh, reduce mod p: $\bar{g}\bar{h} = 0$ in the integral domain $\mathbb{F}_p[x]$, contradiction. For a factorization $f = gh$ over $\mathbb{Q}$, write $g = \lambda g_0$, $h = \mu h_0$ with g_0, h_0 primitive and compare contents.

Hint 2. Eisenstein: if $f = gh$ in $\mathbb{Z}[x]$ with both factors non-constant, reduce mod p: $\bar{f} = \bar{a}_n x^n$, so $\bar{g} = bx^r$, $\bar{h} = cx^s$ with $r, s \geqslant 1$. Then p divides both constant terms of g, h, and $p^2 \mid a_0$. Reduction: a factorization over $\mathbb{Z}$ into factors of positive degree reduces mod p to one with the same degrees (the leading coefficient is not divisible by p).

Hint 3. For $\Phi_p(x) = 1 + x + \cdots + x^{p-1} = \frac{x^p-1}{x-1}$ substitute $x = y + 1$: $\Phi_p(y+1) = \sum_{k=1}^{p} \binom{p}{k} y^{k-1}$, which is Eisenstein at p.

Problem 2

Hint 1. Suppose $(x-1)\cdots(x-n) - 1 = g(x)h(x)$ with $g, h \in \mathbb{Z}[x]$ of smaller degree. Evaluate at $x = 1, \ldots, n$.

Hint 2. $g(i)h(i) = -1$ forces $g(i) = -h(i) = \pm 1$, so $g + h$ has the n roots $1, \ldots, n$ but degree $< n$. Hence $h = -g$.

Hint 3. Then $(x-1)\cdots(x-n) - 1 = -g(x)^2$ has negative leading coefficient, contradiction. Answer: no, for no n.

Problem 3

Note. As printed the statement is false whenever n is not a power of 2: with $Y = X(X+a)$ the polynomial is $Y^{2n} + 1$, and for $n = 2^s m$ with $m > 1$ odd, $Y^{2^{s+1}} + 1$ divides it (e.g. $X^6(X+a)^6 + 1$ is divisible by $X^2(X+a)^2 + 1$). The original IMC 2012 problem has the exponent 2^n: prove that $X^{2^n}(X+a)^{2^n} + 1$ is irreducible. The hints follow this version; write $N = 2^n \geqslant 2$.

Hint 1. Let ε be a root and $\zeta = \varepsilon(\varepsilon + a)$. Then $\zeta^N = -1$, so ζ is a primitive $2N$-th root of unity and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \varphi(2N) = N$. Since $\zeta \in \mathbb{Q}(\varepsilon)$, it suffices to show $[\mathbb{Q}(\varepsilon) : \mathbb{Q}(\zeta)] = 2$.

Hint 2. ε is a root of $X^2 + aX - \zeta$ over $\mathbb{Q}(\zeta)$, so you need: $a^2 + 4\zeta$ is not a square in $\mathbb{Q}(\zeta)$. Take the norm $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}$: a square would have a norm that is the square of a rational number.

Hint 3. The conjugates of ζ are all roots of $T^N + 1$, so $\prod(a^2 + 4\zeta') = 4^N\big((a/2)^{2N} + 1\big)$ (N even). With $a/2 = u/v$ you would need $u^{2N} + v^{2N}$ to be a square, i.e. $x^4 + y^4 = z^2$ with $x = u^{N/2}$, $y = v^{N/2}$.

Hint 4. By Fermat (infinite descent) $x^4 + y^4 = z^2$ has no solutions with $xy \neq 0$. The case $a = 0$ is $X^{2N} + 1 = \Phi_{4N}(X)$, irreducible.

Problem 4

Hint 1. Write $f = ax^2 + bx + c$ and use $k = 0, 1, -1$.

Hint 2. $5 \mid c$, $5 \mid a+b$, $5 \mid a-b$, so $5 \mid 2a$, hence $5 \mid a$ and $5 \mid b$.

Hint 3. General statement: if $\deg f \leqslant p-1$ and $p \mid f(k)$ for all integers k, then p divides all coefficients, because $\bar{f} \in \mathbb{F}_p[x]$ has p roots but degree $< p$. The bound is sharp: $x^p - x$.

Problem 5

Hint 1. Consider $d(x,y) = \dfrac{f(x)-f(y)}{x-y}$, a polynomial in two variables of total degree $\leqslant n-1$. You must show it is integer-valued at all integer points with $x \neq y$; you know this for $0 \leqslant x < y \leqslant n$.

Hint 2. Change variables to x and $z = n-y$. The known points include the triangle $T = \{(x,z) : x, z \geqslant 0,\ x+z \leqslant n-1\}$ (there $x < y$ holds automatically).

Hint 3. Expand $d = \sum_{i+j \leqslant n-1} c_{ij} \binom{x}{i}\binom{z}{j}$. Evaluating on T in increasing order of $i+j$ (or by finite differences in each variable) shows all $c_{ij} \in \mathbb{Z}$, so d is integer-valued on all of $\mathbb{Z}^2$.